INFLUENCER PROFILING SYSTEM USING MACHINE LEARNING AND DEEP LEARNING TECHNIQUES

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Contents

ABSTRACT

The project "Influencer Profiling System Using Machine Learning and Deep Learning Techniques" introduces a multimodal framework for automatically classifying social media influencers based on both textual and visual content from Instagram. As influencer marketing continues to expand, brands face increasing challenges in identifying suitable influencers across diverse niches. This study aims to automate that process by integrating machine learning and deep learning models that jointly analyse captions and images to categorize influencers into domains such as beauty, fashion, fitness, food, and travel.

The original dataset comprised information from over 33,000 influencers and more than one million posts; however, for this study, a representative subset of 1,500 influencers was selected to ensure balanced category representation and efficient model training. Text embeddings were generated using a BERT-based model, while visual embeddings were extracted using InceptionV3. These were combined through an attention-based neural network, which outperformed single-modality approaches. The results confirm that integrating textual and visual cues provides a richer semantic understanding of influencer behaviour and enhances profiling performance.

The study demonstrates that while captions reflect tone and engagement style, images capture lifestyle and thematic consistency, and their fusion leads to superior categorization and interpretability. This work presents a scalable, data-driven approach to influencer profiling, offering actionable insights for digital marketing analytics and enabling brands to identify suitable collaborators and optimize campaign strategies in a rapidly evolving social media landscape.

Keywords:

Influencer Profiling, Multimodal Learning, Deep Learning, Machine Learning, Attention Mechanism, BERT, InceptionV3, Instagram Analytics, Neural Networks, Digital Marketing Automation.

1. INTRODUCTION

1.1 Background of the Project

Influencer marketing has become one of the most powerful strategies in modern digital promotion, leveraging the reach and credibility of individuals on social media platforms to shape audience perceptions and purchasing decisions. Brands increasingly collaborate with influencers to enhance brand awareness, foster customer engagement, and promote products in a more authentic and relatable manner.

Research indicates that consumers tend to trust influencer recommendations more than direct brand advertisements, resulting in a significant shift in marketing investments toward influencer-based strategies. The global influencer marketing industry, valued at approximately USD 2 billion in 2017, was projected to surpass USD 10 billion by 2020, highlighting its rapid expansion and economic importance. It is projected to reach approximately USD 32.55 billion in 2025. Additionally, in 2025, 59% of marketers reported they plan to partner with *more* influencers than in 2024. This growing industry calls for scalable, data-driven systems that can accurately profile and classify influencers across different domains such as fashion, fitness, food, and travel.

1.2 Literature Review

The rise of influencer marketing has prompted extensive academic attention toward automated and data-driven influencer identification. Early studies focused on social and behavioral aspects—Abidin (2016) explored how influencers establish authenticity through personal storytelling, while Casaló et al. (2018) emphasized credibility and audience trust as critical factors in influencer effectiveness. Joshi et al. (2023) provided a comprehensive overview of influencer marketing research, noting the shift from manual influencer discovery to AI-assisted systems. However, these studies relied heavily on engagement metrics or qualitative insights, lacking automated or scalable analytical methods.

Traditional machine learning techniques such as Support Vector Machines (SVM) and Random Forests were initially used for sentiment and user classification (Jain & Sah, 2022). While effective on small datasets, these models required extensive manual feature engineering and failed to capture the multimodal and contextual nature of influencer content. Deep learning models later addressed these limitations by automatically learning hierarchical features from raw data. Transformer-based models such as BERT (Devlin et al., 2019) improved contextual understanding of textual data, while convolutional neural networks (CNNs) like InceptionV3 (Szegedy et al., 2016) and ResNet provided high accuracy in visual content analysis.

A significant advancement came with Kim et al. (2020)**,** who introduced the Multimodal Post-Attentive Profiling (MPAP) framework for influencer marketing. Their model combined textual and visual embeddings using attention mechanisms to classify influencers into categories like beauty, fitness, fashion, and travel. Using a large Instagram dataset of over 33,000 influencers and 10 million posts, they demonstrated that integrating text and image features substantially improved classification performance—achieving over 60% accuracy with image-based models compared to 44% with text alone. The attention-based fusion approach also improved interpretability, setting a strong foundation for later multimodal profiling methods.

Building on this, Khalil et al. (2023) used cross-modal transformers for hashtag prediction, showing the benefits of fusing textual and visual representations. Similarly, Sánchez Villegas et al. (2023) applied multimodal attention models to influencer content on Twitter, confirming the superiority of joint text–image analysis over single-modality approaches. Meanwhile, graph-based models such as Influencer Rank (Kim et al., 2023) and INFLECT-DGNN (Tiukhova et al., 2023) explored influencer discovery through social network relationships, though they focused more on influence propagation than on content-based profiling.

Overall, the literature highlights a clear transition from traditional engagement metrics to advanced multimodal deep learning systems, revealing both progress and existing research gaps.

1.3 Rationale of the Work

Despite the rapid growth of influencer marketing, identifying suitable influencers across various niches remains a major challenge. Traditional selection processes depend on manual curation or limited agency databases, which lack scalability, objectivity, and adaptability. Given the data-rich environment of social media platforms like Instagram, there is a strong need for automated systems capable of analysing large-scale multimodal content—text and images—to generate comprehensive influencer profiles.

Developing such intelligent frameworks can assist brands in selecting appropriate influencers for specific campaigns, improving targeting accuracy, and enhancing overall marketing efficiency. Furthermore, a multimodal approach ensures a holistic understanding of influencer identity by considering both linguistic patterns (captions, hashtags) and visual cues (themes, lifestyle, and aesthetics).

1.4 Contribution of the Work

This project introduces a multimodal deep learning framework that integrates textual and visual features from social media posts to automatically classify influencers into key marketing categories such as beauty, fashion, fitness, food, and travel. From a business standpoint, this approach enables brands and marketing agencies to efficiently identify, segment, and target influencers whose content aligns with specific campaign objectives—reducing the time, cost, and subjectivity associated with manual influencer selection.

The system employs attention-based mechanisms to emphasize posts most representative of an influencer's core domain, improving both interpretability and model precision. Experimental evaluations on a large-scale Instagram dataset—originally comprising over 33,000 influencers and approximately one million posts, from which a subset of 1,500 influencers was used—demonstrated that the integrated model achieved an accuracy of 85%, significantly outperforming models based on text or images alone.

Beyond technical performance, this research has strategic business implications. It contributes a comprehensive influencer dataset that can support data-driven marketing decisions, enabling companies to better understand influencer behaviour, optimize brand collaborations, and improve return on investment (ROI) in influencer campaigns. Overall, the proposed system provides a scalable, interpretable, and automation-ready solution that bridges the gap between advanced AI analytics and practical digital marketing applications.

2. DATA DESCRIPTION

This study utilizes a large-scale Influencer and Brand Dataset originally collected from Instagram, designed to support research in influencer profiling, content analysis, and brand–influencer relationship modelling. The dataset comprises both textual and visual information**,** enabling multimodal analysis through machine learning and deep learning methods.

2.1 Dataset Overview

The dataset contains approximately 1.6 million Instagram posts from 33000 influencers, making it one of the most comprehensive public resources for influencer marketing research. It includes detailed metadata such as captions, hashtags, engagement statistics (likes, comments), timestamps, and visual media files associated with each post.

2.1.1 Dataset Components

The dataset is organized into the following files:

(a) post_info.txt

This file lists all 1,601,074 posts with the following attributes:

Column	Description
Post ID	Unique identifier for each post (ranging from 0 to 1,601,073).
Username	Instagram username of the influencer who uploaded the post.
JSON File Name	Filename of the corresponding JSON file containing detailed post information.
Image File(s)	Filename(s) of associated image(s); multiple images may belong to a single post.

This file serves as the primary index linking post metadata, textual content, and image data.

(b) json_files.zip

This archive contains JSON files corresponding to each post. Each JSON file includes rich post-level metadata such as:

- Caption text
- Number of likes and comments
- Usertags and hashtags

(c) img_fi[xx].zip

The dataset provides 16 zip files containing all image data, with a total of 1,601,074 images**.** All images are resized to uniform dimensions for computational efficiency. These images are used to extract visual embeddings using pre-trained deep learning models (e.g., InceptionV3).

(d) profiles_influencers.zip

This archive contains profile data for 33,000 influencers.
Each profile file (named after the influencer's username) contains tab-separated fields:

Field	**Description**
Name	Influencer's display name
Followers	Number of followers
Followees	Number of accounts followed
Posts	Total number of posts
URL	Link to the influencer's profile
Category	Main content domain (e.g., beauty, fashion, food, travel)
Bio	Profile biography text
E-mail	Contact email (if available)
Profile Picture	Link or reference to profile image

2.1.2 Data Relevance

The dataset's multimodal nature—comprising both textual and visual content—makes it particularly well-suited for multimodal deep learning models.
By linking captions (language) and images (visual cues). Its scale and richness enable robust model training and evaluation, facilitating generalizable and interpretable influencer profiling.

2.2 Data Preprocessing

Data preprocessing plays a crucial role in preparing textual and visual content for feature extraction and classification. The following steps were performed to ensure the dataset was clean, consistent, and suitable for training multimodal deep learning models.

2.2.1 Initial Data Loading and Cleaning

The raw dataset contained information about over 1.6 million posts from more than 33,000 influencers.
Before analysis, the following preprocessing steps were applied:

1. **Conversion of Non-Numeric Values**
   The #Followers column, which occasionally contained non-numeric or string-formatted values, was converted into a numeric data type for consistency.
2. **Filtering by Follower Count and Post Frequency**
   - Only influencers with more than 50,000 followers were retained to ensure sufficient reach and influence.
   - Influencers with fewer than 20 posts were removed to maintain reliability and representativeness in content patterns.
     After filtering, a refined dataset of 1500 active and established influencers was obtained. (here)
3. **Text Cleaning of Captions**
   The caption text was standardized and cleaned using the following techniques:
   - URL removal via regular expressions.
   - Emoji conversion into textual descriptions for semantic retention.
   - Whitespace and newline normalization to remove unnecessary characters.

2.2.2 Sampling Strategy

Due to the large size of the dataset, a representative sample was created for efficient model training and validation.

1. **Rationale for Sampling**
   Training BERT-based and CNN-based models on the full dataset would require excessive computational resources. Sampling allows faster experimentation while maintaining statistical diversity.
2. **Stratified Sampling by Category**
   - A stratified sampling approach was adopted to ensure balanced representation across influencer categories.
   - For each category, up to 170 influencers were selected.
3. **Fixed Number of Posts per Influencer**
   - For each influencer, exactly 20 posts were randomly selected.
   - This ensured a consistent input structure for the attention aggregation model and prevented overrepresentation from highly active influencers.

4. **Final Combined Dataset**
   The sampled subsets were merged into a unified Data Frame of 1500 influencers, ready for feature extraction and modelling.

2.2.3 Text Embedding and Representation

Textual information from influencer captions was converted into dense numerical embeddings using a pre-trained BERT model**.**

1. **Tokenization and Preprocessing**
   Captions were tokenized using BERT-base-multilingual-cased, which converts text into token sequences and generates corresponding attention masks to ignore padding tokens during inference.
2. **Embedding Generation**
   The BERT model produced 768-dimensional embeddings for each post, derived from the [CLS] token representation. These embeddings encode semantic information from captions.
3. **Feature Utilization**
   The resulting embeddings served as the input features for the downstream attention and classification modules, representing the linguistic characteristics of influencer posts.

2.2.4 Image Data Processing

In addition to textual data, the visual content from Instagram posts was used to extract complementary features.

1. **Image Loading and Resizing**
   - A total of 38000 images associated with 29900 posts were loaded.
   - Image dimensions varied across posts, with common sizes including (256×256), (256×320), and (384×256).
   - All images were resized to 299×299 pixels for compatibility with the InceptionV3 architecture.
2. **Feature Extraction using InceptionV3**
   - A pre-trained InceptionV3 model was used, with its final classification layer replaced to output 1024-dimensional embeddings.
   - The convolutional base was frozen to leverage transfer learning, extracting rich visual features from images.
3. **Influencer-Level Embedding Aggregation**
   - For influencers having multiple images per post, image embeddings from their posts were averaged to create a single 1024-dimensional representative visual vector**.**

2.3 Model Architecture

Overview of the Multimodal Framework

The architecture consists of three main modules:

1. **Text Feature Extractor (BERT)** – captures linguistic and contextual meaning from captions.
2. **Image Feature Extractor (InceptionV3)** – encodes visual aesthetics and content features from posts.
3. **Fusion and Classification Module** – combines both modalities to predict influencer categories.

Each influencer is represented by two feature vectors:

- A 768-dimensional text embedding (from BERT CLS token)
- A 1024-dimensional image embedding (from InceptionV3 output)

For consistency, only influencers with at least 20 posts and more than 50,000 followers were included in the dataset. This ensured that each influencer had enough content for meaningful multimodal representation.

3. METHODOLOGY

This study utilizes a combination of machine learning and deep learning techniques to analyze and classify influencers based on their social media content. The integration of both approaches allows for performance comparison between classical algorithms that rely on structured feature representations and advanced deep neural networks that automatically extract semantic and visual information from unstructured data.

After generating embeddings from text captions using BERT and from images using InceptionV3, all posts were grouped by their corresponding usernames to ensure that each influencer was represented as a unified entity. This grouping step aggregates post-level embeddings into influencer-level representations, allowing the models to capture the overall thematic and stylistic consistency of each influencer's content rather than treating individual posts as independent samples. By doing so, the system maintains contextual integrity and enables more accurate classification based on an influencer's collective posting behaviour.

The grouped influencer-level embeddings were then used as inputs to both traditional machine learning classifiers and deep learning architectures for final category prediction. The primary techniques employed in this project are outlined below.

3.1 Machine Learning Techniques

Machine learning (ML) techniques serve as foundational models in this research. They were applied to processed and structured multimodal data to provide benchmark results for influencer classification. These algorithms rely on embeddings extracted from captions and images and help evaluate the added benefit of deep learning and multimodal integration.

For the baseline machine learning models, the text and image embeddings of all posts belonging to the same influencer were averaged to form a single fixed-length vector. This approach treats all posts as equally important, without distinguishing between representative and irrelevant content.

(a) Random Forest

Random Forest is an ensemble-based machine learning method that improves classification accuracy through aggregation of multiple decision trees. Each tree in the forest is trained on a bootstrapped subset of the data, and random feature selection is applied at each split to reduce correlation among trees. The final classification is determined by a majority vote among all trees.

Mathematically, if $T_1, T_2, ..., T_k$ are decision trees, the final prediction $\hat{y}$ is given as:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), ..., T_k(x)\}$$

Each decision tree minimizes impurity at each split, typically using Gini Impurity (G) or Entropy (H):

$$G = 1 - \Sigma(p_i^2)$$
$$H = -\Sigma(p_i \log_2(p_i))$$

where p_i represents the proportion of samples belonging to class i.

Random Forests are known for their robustness against overfitting and their ability to handle high-dimensional data. In the context of influencer profiling, the model captured non-linear relationships between image and text features, effectively distinguishing influencers across domains such as beauty, fashion, and fitness.

(b) Naive Bayes

Gaussian Naïve Bayes (GNB) is a probabilistic classifier grounded in Bayes' theorem, assuming independence between features. It calculates the posterior probability for each class C_k given feature vector $x = (x_1, x_2, ..., x_n)$ as:

$$P(C_k \mid x) = [P(x \mid C_k) * P(C_k)] / P(x)$$

Under the Gaussian assumption, the conditional probability for each feature is modeled as:

$$P(x \mid C_k) = \Pi\ (1 / \sqrt{(2\pi\sigma_{ki}^2)}) * \exp(-(x_i - \mu_{ki})^2 / (2\sigma_{ki}^2))$$

where μ_{ki} and σ_{ki}^2 denote the mean and variance of feature x_i for class C_k.

GNB is computationally efficient and performs surprisingly well even when feature independence assumptions are partially violated. In this research, GNB provided a quick baseline for evaluating how distinct influencer categories are distributed in the embedding space.

(c) K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is an instance-based, non-parametric classification method that relies on proximity between data points. It assigns a class to a new input sample based on the majority vote of its k closest neighbors in the feature space.

The most common metric for measuring similarity is the Euclidean distance:

$$d(x_i, x_j) = \sqrt{\Sigma(x_{in} - x_{jn})^2}$$

The predicted class for a new sample x is: $\hat{y} = \text{mode}(y_i)$ for the k nearest neighbors of x.

KNN does not involve any training phase, making it simple but computationally expensive for large datasets. In this project, KNN grouped influencers whose multimodal embeddings were similar, providing insight into natural content clusters such as beauty–fashion overlap or food–travel intersections.

(d) Support Vector Machine (SVM)

Support Vector Machine (SVM) is a margin-based classifier that seeks the optimal hyperplane separating classes with the maximum possible margin.

For training data (x_i, y_i), where $y_i \in \{-1, +1\}$, SVM solves the optimization problem:

$$\text{minimize } (1/2)||w||^2 \text{ subject to } y_i(w \cdot x_i + b) \geq 1$$

The decision boundary is defined by the function:

$$f(x) = \text{sign}(w \cdot x + b)$$

In scenarios where data is not linearly separable, kernel functions are employed to project data into higher dimensions. The most common kernel used is the Radial Basis Function (RBF):

$$K(x_i, x_j) = \exp(-\gamma||x_i - x_j||^2)$$

Here, γ controls the influence of a single training example. SVM excels at finding clear class boundaries even in complex feature spaces. For influencer profiling, SVM proved highly effective in separating visually and semantically similar influencer types (e.g., lifestyle vs. travel).

3.2 Deep Learning Techniques

3.2.1 Attention-Based Neural Network for Influencer Representation

Once both caption-based embeddings (from BERT) and image-based embeddings (from InceptionV3) are concatenated for each post, multiple such multimodal vectors exist for every influencer, since influencers generally create several posts over time. However, not all posts are equally informative or representative of the influencer's identity. Some posts may be promotional, off-topic, or stylistically inconsistent with the influencer's usual theme. Treating all posts equally could therefore dilute the quality of the final influencer representation.

To address this, the system employs an Attention-Based Neural Network that learns to assign importance weights to each post. The goal of the attention mechanism is to determine which posts best reflect the influencer's true category and should therefore have a stronger influence on the final decision.

Role of Attention

The attention mechanism takes as input the set of multimodal embeddings corresponding to each influencer and calculates a relevance score for each embedding. These scores represent how well each post aligns with the influencer's dominant content style and category characteristics. Posts that strongly exhibit thematic consistency—such as similar visual aesthetics, subject matter, caption wording style, and emotional tone—receive higher attention weights. Conversely:

- Posts showing irregular or off-topic content
- Posts with weak or ambiguous category cues
- Posts with low semantic or visual distinctiveness

are assigned lower weights, reducing their effect on the influencer's final representation.

Computation of the Aggregated Embedding

Once attention weights are computed, they are applied to the multimodal embeddings through a weighted sum operation:

$$\text{Influencer Representation} = \sum_{i=1}^{n} \alpha_i \cdot \text{Embedding}_i$$

where:

- Embedding_irepresents the multimodal feature vector for the i^{th}post
- $\alpha_i$is the attention weight assigned to that post
- and n is the total number of posts for the influencer.

This results in a single fixed-length embedding vector that encapsulates both textual behaviour and visual style in a balanced and context-aware manner, rather than simply averaging all posts or selecting posts manually.

Neural Classification Layer

After the attention mechanism aggregates the most representative multimodal features for each influencer, the resulting influencer-level embedding is forwarded to the Neural Classification Layer. This layer serves as the final decision-making component of the system, responsible for mapping the learned embedding to one of the predefined influencer categories (such as fashion, beauty, fitness, travel, etc.).

The classification module is implemented as a fully connected feed-forward neural network. The aggregated embedding vector is first passed through one or more dense layers that apply linear transformation followed by non-linear activation functions (such as ReLU). These transformations enable the network to learn and refine higher-order relationships between semantic text cues and visual patterns present within the influencer's content.

At the final stage, the output from the last dense layer is fed into a softmax activation layer, which converts the raw network outputs (logits) into a probability distribution over all influencer categories. The category with the highest probability is selected as the predicted profile class for that influencer. This probabilistic formulation also allows interpretability in terms of how strongly the model associates an influencer with each category.

To ensure that the classifier learns effectively, the network parameters are optimized using the Cross-Entropy Loss function. Cross-Entropy is well-suited for multi-class classification tasks, as it penalizes confident but incorrect predictions more heavily, guiding the model to improve discrimination between similar influencer categories. The optimization is performed using the Adam Optimizer, which adapts learning rates dynamically for individual parameters, leading to faster and more stable convergence compared to traditional gradient descent.

Additionally, to prevent the model from overfitting (memorizing training influencers instead of learning general category characteristics), an Early Stopping strategy is employed. During training, the model's performance is continuously monitored on a validation set. If performance stops improving for a defined number of epochs, training is halted automatically. This ensures that the final classifier balances both learning capacity and generalization to unseen influencers.

Hyperparameter Tuning:

To further enhance model robustness and accuracy, an extensive hyperparameter tuning phase was carried out, exploring over 250 different parameter combinations**.** The tuning process systematically varied critical parameters such as learning rate, batch size, number of hidden layers, dropout rate, activation functions, and optimizer configurations (Adam, SGD). Both grid search and randomized search techniques were used to identify the most effective combination that maximized validation accuracy while minimizing overfitting. The final tuned configuration—consisting of a learning rate of 1e-3 and dropout rate of 0.3—achieved the highest validation accuracy of 85%. This structured optimization process was instrumental in stabilizing training and achieving superior classification performance within the multimodal influencer profiling framework.

By integrating these mechanisms, the neural classification layer effectively translates the learned multimodal representation into a reliable and category-consistent influencer identity prediction, completing the profiling pipeline.

4. RESULTS AND DISCUSSION

This section presents the experimental results obtained from various machine learning and deep learning models applied to the influencer profiling task. The evaluation was performed using text-only, image-only, and combined (text + image) feature inputs. Accuracy was used as the primary performance metric to assess the classification capability of each model in predicting influencer categories.

4.1 Overall Model Performance

Model	Input Modality	Accuracy
Random Forest	Text	45%
	Image	73.33%
	Text + Image	**75%**
K-Nearest Neighbors (KNN)	Text	39%
	Image	58%
	Text + Image	**74%**
Support Vector Machine (SVM)	Text	51%
	Image	78%
	Text + Image	**83%**
Gaussian Naïve Bayes	Text	27.67%
	Image	65%
	Text + Image	**76.33%**
Neural Classifier (Deep Learning)	Text	56%
	Image	79%
	Text + Image	**85%**

4.2 Analysis of Text-based Models

When trained solely on textual data (captions), the performance of classical machine learning models remained relatively modest.

- The Support Vector Machine (SVM) achieved the highest text-based accuracy (51%), followed by the Neural Classifier (56%), indicating that deep learning models were able to capture richer contextual representations from text embeddings produced by BERT.
- Models like Random Forest and KNN performed lower (45% and 39%, respectively), reflecting their limited capacity to understand the semantic nuances and contextual dependencies in influencer captions.
- Naïve Bayes, which assumes feature independence, performed the weakest (27.67%), highlighting the challenge of applying simple probabilistic models to social media text, where word dependencies and contextual cues are strong.

Overall, text alone provides limited discriminative power for influencer profiling, as captions are often short, stylistically diverse, and contain non-standard expressions, emojis, or hashtags that make pure linguistic analysis difficult.

4.3 Analysis of Image-based Models

When models were trained on image features extracted using the InceptionV3 network, there was a substantial improvement in performance across all techniques.

- The Neural Classifier achieved an image-based accuracy of 79%, followed closely by SVM at 78% and Random Forest at 73.33%**.**
- Even simpler models such as KNN (58%) and Naïve Bayes (65%) performed significantly better compared to their text-only versions.

These results indicate that visual features are more informative than textual ones for influencer profiling. Images provide strong contextual signals about the influencer's domain — for instance, fitness posts often include gym equipment, while fashion influencers post outfit and brand-related images.

The success of image-based approaches also highlights the importance of transfer learning with pre-trained models like InceptionV3, which can effectively generalize visual representations to new domains with limited fine-tuning.

4.4 Multimodal (Text + Image) Fusion Results

The integration of text and image embeddings led to the highest classification accuracies across all models, demonstrating the effectiveness of multimodal fusion.

- The Neural Classifier achieved the best overall performance with 85% accuracy**,** outperforming all traditional algorithms.

- Among machine learning models, Random Forest (75%) and SVM (83%) performed competitively, indicating their robustness when provided with rich, multimodal features.
- Even simpler models like KNN (74%) and Naïve Bayes (76.33%) benefited from multimodal input, showing the complementary nature of text and visual cues.

These findings validate that combining textual and visual modalities provides a more holistic representation of influencers. While captions convey semantic intent (e.g., promotions, brand partnerships), images visually express the influencer's style, theme, and product focus. The attention-based neural model effectively leveraged both modalities to emphasize the most representative posts, leading to superior classification performance.

4.5 Discussion

The experimental results clearly demonstrate three major trends:

1. Deep learning models outperform traditional machine learning approaches**,** owing to their ability to learn complex, non-linear relationships between multimodal features.
2. Image features are more discriminative than textual features for influencer classification, emphasizing the visual nature of social media platforms like Instagram.
3. Multimodal fusion provides the highest accuracy (85%)**,** confirming that integrating both caption and image information produces a more comprehensive understanding of influencer profiles.

The superior performance of the Neural Classifier with Attention highlights its ability to dynamically focus on relevant posts, reducing noise from irrelevant or off-topic content. Furthermore, the use of pre-trained models (BERT and InceptionV3) effectively reduced training time while enhancing accuracy through transfer learning.

5. SUMMARY AND CONCLUSIONS

1. This study aimed to develop an automated and data-driven system for influencer profiling using multimodal deep learning techniques. By leveraging both textual and visual content from social media posts, the project sought to accurately classify influencers into predefined categories such as fashion, fitness, beauty, food, and travel.
2. The research began with comprehensive data preprocessing, which involved filtering influencers based on follower count and post activity, cleaning textual data, handling missing values, and standardizing image inputs. Text embeddings were generated using the BERT-base-multilingual-cased model, while visual embeddings were extracted from InceptionV3, creating rich representations of both modalities. The final multimodal dataset enabled comparative experiments between classical machine learning models and deep learning approaches.
3. The results revealed that image features carried more discriminative power than textual ones, reflecting the visual-centric nature of influencer marketing platforms like Instagram. However, the integration of both modalities consistently yielded the best outcomes. Among all evaluated models, the Neural Classifier with an attention mechanism achieved the highest classification accuracy of 85%, outperforming traditional algorithms such as Random Forest (75%) and SVM (83%). The attention mechanism enabled the model to focus on the most relevant posts, enhancing interpretability and reducing noise from irrelevant content.
4. The findings demonstrate that multimodal fusion significantly enhances influencer classification performance, validating the importance of combining linguistic and visual cues in computational social media analysis. Deep learning models, particularly those leveraging transfer learning, proved to be more adaptable and effective in capturing the complex relationships between influencers' content and their domains.

5.1 Key Contributions

- Developed a multimodal influencer profiling framework integrating BERT-based text embeddings and InceptionV3-based image features.
- Demonstrated that deep learning models outperform traditional machine learning techniques in influencer classification tasks.
- Achieved an accuracy of 85% using an attention-based neural classifier on a large-scale influencer dataset.
- Provided a balanced and scalable dataset through stratified sampling and fixed post selection per influencer.

5.2 Future Work

While the proposed system achieved high accuracy and interpretability, several avenues for future improvement remain:

- **Extension to Video and Audio Content:** Incorporating multimodal inputs from short-form videos (e.g., Instagram Reels, TikTok) could provide richer behavioural signals.
- **Incorporation of Engagement Metrics:** Features such as likes, comments, and follower growth rates could enhance influencer ranking and credibility assessment.
- **Use of Advanced Multimodal Transformers:** Emerging models like CLIP**,** ViLT, or LLaVA can be explored for improved joint understanding of visual and textual content.
- **Explainability and Bias Analysis:** Future studies could investigate interpretability frameworks to better understand model decisions and reduce potential category bias.

5.3 Final Remarks

Overall, this research demonstrates the potential of multimodal deep learning to revolutionize influencer marketing analytics. By integrating advanced text and image understanding methods, the proposed system provides a scalable, accurate, and intelligent solution for automated influencer profiling — bridging the gap between computational analysis and real-world marketing applications.

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